

# YUNSUNG CHUNG

213 999 8640 ◇ cys1102@gmail.com ◇ New Orleans, Louisiana

[Website](#) | [LinkedIn](#) | [Google Scholar](#)

## SUMMARY

---

Ph.D. student in Computer Science at Tulane University working on trustworthy medical AI, including clinical world models for dynamic decision support, reward-aligned diffusion models, and privacy–utility evaluation of synthetic data. Publications at USENIX Security, MICCAI, ICASSP, and AI-ML Systems; research internship at Samsung Research America.

## EDUCATION

---

|                                                                                           |                            |
|-------------------------------------------------------------------------------------------|----------------------------|
| <b>Tulane University</b><br>Ph.D. in Computer Science                                     | <i>Aug 2022 – Present</i>  |
| <b>The Australian National University</b><br>M.S. in Computer Science                     | <i>Jul 2019 – Jul 2021</i> |
| <b>Dongguk University</b><br>B.S. in Electronics, Information & Communication Engineering | <i>Mar 2012 – Feb 2018</i> |

## RESEARCH INTERESTS

---

- Diffusion model finetuning and reward-aligned generation under domain-specific constraints.
- Privacy–utility analysis and trustworthy synthetic data generation.
- World models and latent dynamics for sequential forecasting and counterfactual reasoning.
- Representation learning under distribution shift, label scarcity, and noisy supervision.

## EXPERIENCE

---

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>Samsung Research America – Research Intern</b><br>Mountain View, California                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <i>May 2025 – Aug 2025</i><br>Supervisor: Dr. Sharanya Desai |
| <ul style="list-style-type: none"><li>• Designed temporal-aware loss functions for learning from large-scale time-series data with sparse and irregularly sampled labels.</li><li>• Built end-to-end deep learning pipelines for multi-modal sensor data, from preprocessing to deployment-ready evaluation.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                              |
| <b>Tulane University (TRIAD Lab) – Research Assistant</b><br>New Orleans, Louisiana                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Aug 2022 – Present</i><br>Supervisor: Dr. Jihun Hamm      |
| <ul style="list-style-type: none"><li>• Developed a reward-aligned finetuning framework for diffusion models using VLMs as differentiable critics, achieving SOTA on domain-specific image synthesis benchmarks.</li><li>• Authored a comprehensive survey on synthetic data utility and privacy, systematizing 130+ papers and accepted at a top-tier security venue.</li><li>• Built latent-space world models with intervention conditioning for sequential forecasting, supporting anytime prediction and counterfactual simulation.</li><li>• Designed contrastive and semi-supervised learning methods for segmentation under extreme label scarcity (5–10% labeled data).</li><li>• Implemented few-shot augmentation pipelines using diffusion inversion and distribution matching, improving downstream classifier accuracy by 5–12%.</li></ul> |                                                              |
| <b>Airport Railroad Express – Hardware Engineer</b><br>Incheon, South Korea                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <i>Nov 2017 – May 2019</i>                                   |

- Analyzed and tested embedded hardware systems for automated fare collection and real-time passenger service infrastructure.

## PUBLICATIONS

---

### Peer-Reviewed Conferences & Workshops

- [1] **Yunsung Chung**, Keum San Chung, Migyeong Gwak, Han Feng, Yingshuo Liu, Chanh Lim, Viswanathan, Nassir Marrouche, and Sharanya Arcot Desai. *Weighted Temporal Decay Loss For Learning Wearable PPG Data With Sparse Clinical Labels*. **ICASSP**, 2026.
- [2] **Yunsung Chung**, Chanh Lim, Ghassan Bidaoui, Christian Massad, Nassir Marrouche, and Jihun Hamm. *SOFA: Deep Learning Framework for Simulating and Optimizing Atrial Fibrillation Ablation*. **MICCAI**, 2025.
- [3] **Yunsung Chung**, Yunbei Zhang, Nassir Marrouche, and Jihun Hamm. *SoK: Can Synthetic Images Replace Real Data? A Survey of Utility and Privacy of Synthetic Image Generation*. **USENIX Security**, 2025.
- [4] **Yunsung Chung**, Janet Wang, and Jihun Hamm. *Bridging the Gap: Synthetic Data Augmentation through Inversion and Distribution Matching for Few-shot Learning*. **AI-ML Systems**, 2024.
- [5] Janet Wang, **Yunsung Chung**, Zhengming Ding, and Jihun Hamm. *From Majority to Minority: A Diffusion-based Augmentation for Underrepresented Groups in Skin Lesion Analysis*. **MICCAI Workshop**, 2024.
- [6] **Yunsung Chung**, Chanh Lim, Chao Huang, Nassir Marrouche, and Jihun Hamm. *FBA-Net: Fore-ground and Background Aware Contrastive Learning for Semi-Supervised Atrium Segmentation*. **MICCAI Workshop**, 2023.

### Manuscripts Under Review

- [7] **Yunsung Chung**, Han Feng, Nassir Marrouche, and Jihun Hamm. *Auditing Long-Horizon Patient Trajectory Modeling in EHR Foundation Model Benchmarks*. Under review at **NeurIPS**, 2026.
- [8] **Yunsung Chung**, Alex El Darzi, Carlo El Khoury, Han Feng, Nassir Marrouche, and Jihun Hamm. *CRAFT: Clinical Reward-Aligned Finetuning for Medical Image Synthesis*. Under review at **NeurIPS**, 2026.
- [9] **Yunsung Chung**, Yingshuo Liu, Abboud F Hassan, Han Feng, Mary M Maleckar, Nassir Marrouche, and Jihun Hamm. *Intervention-Aware Clinical World Model for Post-OP Outcome Forecasting in Cardiology*. Under review at **MICCAI**, 2026.

## AWARDS, SERVICE & OUTREACH

---

**Academic Service:** Reviewer, ICML 2026; MICCAI 2024–2026; IEEE TCSVT, 2023.

**Outreach:** College Outreach Volunteer, NeurIPS 2023.

**Honors:** FIRA RoboWorld Cup 2016 – 1st Place, Basketball; International Robot Contest 2016 – 2nd Place, Marathon; First-Class Academic Excellence Scholarship, Dongguk University, 2012, 2015–2017.

## SELECTED SKILLS

---

**ML:** Diffusion models, world models, representation learning, reward alignment, semi-supervised learning, privacy–utility evaluation.

**Data/Tools:** Medical images, time-series, multi-modal sensors, sequential clinical events; Python, PyTorch, HuggingFace, Scikit-learn, AWS, Weights & Biases, SQL.

**Languages:** English, Korean.